25901052 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION November 2025

First SEMESTER

Skill Development Components (SDC) - MG1SDCREM101 - FUNDAMENTALS OF RENEWABLE ENERGY TECHNOLOGY

(2025 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any **15** questions

Each question carries **1** marks

1. Recall one primary energy source. [K] / [CO1]
2. Identify the cause of wind energy. [K] / [CO1]
3. State the main source of fossil fuels. [K] / [CO1]
4. Define the greenhouse effect. [K] / [CO1]
5. Identify the physical principle involved in converting solar radiation into heat. [U] / [CO2]
6. Inspect the role of the reflective surface in a concentrator. [An] / [CO4]
7. Outline gases evolved during dry-process conversion. [An] / [CO4]
8. Inspect the importance of glazing transmittance on heat gain. [An] / [CO4]
9. Implement the factor that increases drag force on a body moving through air [A] / [CO3]
10. Execute the main function of the rotor blades in a wind energy system [A] / [CO3]
11. Categorize hybrid systems that integrate wind and solar. [An] / [CO4]

12. Compare the roles of the rotor blades and the generator in a wind energy system [An] / [CO4]

13. Differentiate between energy crops and municipal solid waste. [An] / [CO4]

14. Compare batch and continuous digestion systems. [An] / [CO4]

15. Examine potential of algae-based bioenergy systems. [An] / [CO4]

16. Examine how biomass energy supports rural and local economies. [An] / [CO4]

17. Inspect methanogenic bacteria roles in digestion. [An] / [CO4]

[1x15 = 15]

Part B

Short Essay

Answer any **3** questions

Each question carries **5** marks

18. Recall the main characteristics of coal, petroleum, and natural gas as conventional energy source [K] / [CO1]

19. Breakdown the construction of a flat-plate collector and inspect how each component contributes to heat absorption. [An] / [CO4]

20. Examine the site selection considerations for a windmill. [An] / [CO4]

21. Compare thermochemical and biochemical conversion processes. [An] / [CO4]

22. Categorize organic materials based on their biogas yield potential. [An] / [CO4]

[5x3 = 15]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

23. List and show the advantages of non-conventional energy sources for sustainable energy development. [K] / [CO1]

24. Compare the types of solar photovoltaic systems. [An] / [CO4]

25. Outline various applications of wind energy in industrial and rural contexts. [An] / [CO4]

26. Examine how biomass can be converted into different forms of energy (solid, liquid, and gas) and outline the advantages of each method [An] / [CO4]

[10x2 = 20]